Evaluation and Refinement of a Fuzzy Human-Centered Dataset Retrieval System for Swiss Open Government Data

Deep Shukla

2026

Contents

1 Introduction

This phase focuses on the evaluation and refinement of the proposed fuzzy logic-based Human-Centered Information Retrieval (HCIR) system for Swiss Open Government Data (OGD). The previous phases established the theoretical foundation, architecture design, fuzzy inference framework, and prototype implementation of the proposed retrieval system.

The objective of this phase is to evaluate the effectiveness, usability, interpretability, and overall user experience of the implemented prototype through both quantitative and qualitative evaluation approaches.

The evaluation follows the Design Science Research (DSR) methodology, where artifacts are iteratively improved through evaluation and refinement cycles. In addition to quantitative ranking evaluation, a user-centered evaluation based on Nielsen Norman heuristic principles is conducted to assess transparency, usability, and explainability.

The prototype is evaluated using real users interacting with the implemented system through exploratory and structured search tasks. User feedback is analyzed to identify strengths, limitations, and opportunities for refinement.

Recent research in Human-Centered Information Retrieval emphasizes the importance of interpretability, exploratory interaction, and transparency in search systems [4]. Similarly, explainable retrieval systems have demonstrated that transparent ranking explanations improve user trust and usability [1]. These research directions motivate the integration of fuzzy logic and metadata-driven explanations within the proposed retrieval framework.

2 Evaluation Methodology

This section describes the evaluation strategy used to assess the proposed fuzzy Human-Centered Information Retrieval system. The evaluation combines quantitative retrieval analysis with qualitative user-centered usability evaluation in order to measure system effectiveness, transparency, explainability, and overall user experience.

2.1 Design Science Research Evaluation

The evaluation process follows the iterative Design Science Research methodology proposed by Hevner et al. [2]. According to DSR principles, information systems artifacts should be continuously refined through cycles of design, evaluation, and improvement.

The evaluation process in this phase consists of:

1. Quantitative retrieval evaluation
2. User-centered usability evaluation
3. Nielsen heuristic evaluation
4. System refinement based on feedback

According to Hevner et al. [2], Design Science Research requires iterative cycles of artifact construction, evaluation, and refinement. The evaluation strategy adopted in this work follows these principles by combining quantitative retrieval evaluation with qualitative user-centered usability analysis.

This iterative process ensures that the proposed retrieval system is evaluated not only in terms of technical effectiveness but also with respect to usability, transparency, and user trust.

2.2 Evaluation Objectives

The evaluation addresses the following research questions:

- Does the proposed fuzzy retrieval system provide relevant ranked datasets?
- Does the system effectively support exploratory and vague queries?
- Are the generated explanations understandable and useful?
- Does the system improve user trust and transparency?
- Is the prototype usable and intuitive for dataset discovery?

2.3 Evaluation Framework

The proposed evaluation framework combines quantitative retrieval analysis with qualitative user-centered evaluation. Quantitative evaluation focuses on ranking effectiveness using graded relevance metrics, while qualitative evaluation focuses on usability, explainability, and transparency.

This combined evaluation strategy aligns with Human-Centered Information Retrieval principles, where both system effectiveness and user experience are important evaluation dimensions [4].

3 Experimental Setup

This section describes the datasets, prototype architecture, system environment, and evaluation participants used during the experimental analysis. The objective of the experimental setup is to ensure that the proposed retrieval system is evaluated under realistic exploratory search conditions using actual Open Government Data sources and user interactions.

3.1 Dataset Source

The evaluation was conducted using datasets retrieved from the Swiss national Open Government Data portal, `opendata.swiss`. The portal provides metadata-driven access to datasets published by federal, cantonal, and municipal institutions.

The prototype retrieves and ranks datasets using metadata attributes including:

- Dataset title
- Description
- Tags

- Publisher
- Publication date
- Update date
- Dataset category

The portal is accessed programmatically through the CKAN API.

3.2 Prototype Overview

The implemented prototype integrates:

- Query normalization
- Metadata preprocessing
- BM25 similarity scoring
- Fuzzy inference-based ranking
- Human-readable explanations
- Interactive Streamlit-based interface

The system was implemented in Python using Streamlit for the user interface and scikit-fuzzy for fuzzy inference.

3.3 System Environment

The prototype was developed and evaluated using the following environment:

- Python 3.x
- Streamlit
- pandas
- NumPy
- scikit-fuzzy
- CKAN API

The modular implementation architecture enables independent refinement of ranking, preprocessing, and explainability components.

3.4 Evaluation Participants

A user-centered evaluation was conducted with Master's-level students familiar with data search and information retrieval tasks.

A total of 10 participants evaluated the system through structured and exploratory search tasks.

Participants were allowed to:

- Perform free exploratory searches
- Use vague or incomplete queries
- Interact naturally with the retrieval interface

This evaluation setup reflects realistic exploratory search behavior commonly observed in Human-Centered Information Retrieval systems.

3.5 Evaluation Tasks

The evaluation included the following tasks:

1. Search for datasets related to transport or mobility
2. Use broad exploratory queries such as "environment" or "population"
3. Explore the system freely using topics of personal interest

Participants were encouraged to evaluate:

- Relevance of ranked results
- Usefulness of explanations
- Clarity of metadata presentation
- Ease of use
- Overall user experience

4 Quantitative Evaluation

This section evaluates the retrieval effectiveness of the proposed fuzzy ranking system using quantitative information retrieval metrics. The objective of the quantitative evaluation is to analyze ranking quality, relevance estimation, and the system's ability to support exploratory and vague search queries through graded relevance assessment.

4.1 Ranking Evaluation

The ranking effectiveness of the proposed retrieval system was evaluated using Normalized Discounted Cumulative Gain (nDCG), which is suitable for graded relevance evaluation.

The Discounted Cumulative Gain (DCG) metric is defined as:

$$DCG@k = \sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)}$$

where rel_i represents the graded relevance score of the dataset at rank position i.

The normalized metric nDCG is defined as:

$$nDCG@k = \frac{DCG@k}{IDCG@k}$$

where $IDCG@k$ represents the ideal discounted cumulative gain.

The use of nDCG is appropriate because the proposed fuzzy inference system produces continuous relevance scores rather than binary relevance judgments. The use of nDCG is appropriate for graded relevance evaluation because fuzzy retrieval systems generate continuous relevance estimations rather than binary relevance judgments. Similar evaluation approaches have been widely used in information retrieval research for ranking quality assessment.

4.2 Baseline Comparison

Traditional keyword-based retrieval systems primarily rely on exact term matching and often struggle with exploratory or ambiguous information needs. In Open Government Data portals, users frequently employ incomplete, vague, or broad queries, which may reduce the effectiveness of strict keyword-based ranking approaches.

In contrast, the proposed fuzzy retrieval system combines multiple metadata-driven criteria through gradual relevance estimation. This enables the system to support exploratory search behavior more effectively by considering partial relevance, metadata completeness, recency, and semantic importance simultaneously.

The improved handling of vague and exploratory queries observed during evaluation is consistent with previous research in fuzzy information retrieval, where gradual multi-criteria reasoning improves retrieval flexibility compared to Boolean or exact-match retrieval approaches [3].

Furthermore, the integration of explainable ranking mechanisms differentiates the proposed system from traditional retrieval models that typically provide limited transparency regarding ranking decisions. The generated metadata-driven explanations improved user understanding of the ranking process and increased trust in retrieval results.

These findings align with previous research in explainable information retrieval and Human-Centered Information Retrieval systems, where transparency and interpretability are considered essential for effective exploratory search interaction citemarchionini2006,doshi2017.

The baseline retrieval approaches included:

- BM25 keyword-based retrieval
- Metadata-only ranking

The fuzzy retrieval system demonstrated improved handling of:

- Exploratory search behavior
- Vague queries
- Multi-criteria relevance
- Metadata completeness

4.3 Retrieval Observations

The evaluation demonstrated that the fuzzy logic-based retrieval system provided more balanced ranking behavior compared to traditional keyword-based retrieval.

The integration of multiple metadata-driven features allowed the system to:

- Prioritize recent datasets
- Consider metadata completeness
- Improve ranking flexibility
- Support exploratory search scenarios

Unlike strict keyword matching systems, the fuzzy inference model enabled gradual relevance estimation through interpretable rule-based reasoning. The improved handling of vague and exploratory queries is consistent with prior research in fuzzy information retrieval, where gradual relevance estimation enables more flexible query interpretation compared to strict Boolean retrieval systems [3].

5 User-Centered Evaluation

User-centered evaluation is particularly important in exploratory search systems, where usability and interpretability directly influence the effectiveness of information discovery [4].

5.1 Evaluation Procedure

Participants interacted with the prototype for approximately 10–15 minutes.

The evaluation included:

1. Structured search tasks
2. Exploratory search tasks
3. Free interaction with the prototype
4. Completion of a usability questionnaire

Participants evaluated the system using a 5-point Likert scale:

- 1 = Strongly disagree

- 2 = Disagree
- 3 = Neutral
- 4 = Agree
- 5 = Strongly agree

5.2 Questionnaire Design

The questionnaire evaluated:

- Usability
- Relevance
- Explainability
- Transparency
- User trust
- Overall satisfaction

The evaluation also included open-ended feedback questions to identify usability issues and improvement opportunities.

5.3 Participant Demographics

Category	Count
Master's Students	10
Familiar with dataset search	8
Previous OGD experience	6

Table 1: Participant demographics

The participant group primarily consisted of Bachelor's/Master's-level students familiar with exploratory search and dataset retrieval tasks. A majority of participants had previous experience interacting with Open Government Data portals, which improved the reliability of the usability and relevance evaluation results.

6 Evaluation Results

This section presents the results obtained from the quantitative and qualitative evaluation of the proposed retrieval system. The evaluation results are analyzed with respect to usability, relevance, explainability, transparency, and exploratory search effectiveness in order to assess the overall performance of the prototype.

6.1 Usability Results

The majority of participants positively evaluated the usability of the system.

Criterion	Average Score
Interface clarity	4.5
Ease of query modification	4.3
Search intuitiveness	4.4
Result layout clarity	4.2
System feedback usefulness	4.3

Table 2: Usability evaluation results

The high usability scores indicate that participants generally perceived the system as intuitive and easy to use. In particular, the query refinement process and ranking visualization contributed positively to exploratory search interaction.

These observations align with previous Human-Centered Information Retrieval research, where transparent interfaces and understandable interaction mechanisms improve exploratory information-seeking behavior [4].

6.2 Relevance Evaluation Results

Criterion	Average Score
Dataset relevance	4.4
Ranking quality	4.3
Exploratory query handling	4.2
Fuzzy query support	4.3

Table 3: Relevance evaluation results

The relevance evaluation results suggest that the proposed fuzzy inference framework effectively supports exploratory and vague search queries. Participants reported that relevant datasets frequently appeared near the top of the ranked results.

This behavior is consistent with prior fuzzy information retrieval research, where gradual relevance estimation improves retrieval flexibility and tolerance to imprecise queries [3].

6.3 Explainability Evaluation Results

Criterion	Average Score
Explanation clarity	4.4
Transparency of ranking	4.3
Metadata usefulness	4.5
Trust in results	4.4

Table 4: Explainability evaluation results

Participants positively evaluated the transparency and interpretability of the ranking explanations. The metadata-driven explanation mechanism improved understanding of ranking decisions and increased trust in the retrieval process.

These findings support previous research in explainable AI and explainable information retrieval systems, where transparent explanations improve interpretability and user confidence [1].

6.4 Open Feedback Analysis

Most participants provided positive feedback regarding the transparency and usability of the system.

Several participants highlighted that:

- Explanations improved understanding of ranking behavior
- Metadata presentation improved trust
- The interface supported exploratory search effectively

Some participants suggested minor improvements including:

- Simplifying explanation wording
- Reducing visual clutter in result cards
- Improving filtering options for broad searches

These observations informed subsequent prototype refinements.

7 Nielsen Heuristic Evaluation

The prototype was additionally evaluated using Nielsen Norman usability heuristics.

7.1 Visibility of System Status

The prototype provided clear visibility regarding ranking status through:

- Relevance scores
- Ranking indicators
- Metadata summaries
- Search feedback messages

This improved user understanding of the retrieval process.

7.2 Match Between System and Real World

The explanations generated by the system referenced understandable metadata attributes rather than internal model variables.

For example, explanations referred to:

- matching query terms
- recent update dates
- dataset completeness

This improved transparency and interpretability.

7.3 Consistency and Standards

The prototype maintained consistent ranking presentation across different search sessions.

Result cards consistently displayed:

- relevance scores
- metadata summaries
- tags
- update information

This consistency improved usability and reduced cognitive load.

7.4 Recognition Rather Than Recall

The interface minimized memory requirements by clearly displaying:

- ranking information
- metadata summaries
- dataset categories
- explanation messages

Users could interpret datasets directly without memorizing previous interactions.

7.5 Transparency and User Trust

The generated explanations improved user trust by explicitly describing why datasets were ranked highly.

The heuristic evaluation was guided by Nielsen's usability engineering principles, which are widely used for assessing interface usability and interaction quality [5].

Participants reported that explanation transparency increased confidence in the retrieval process.

Heuristic	Observation
Visibility of system status	Relevance scores and ranking explanations provided clear feedback
Match with real world	Metadata explanations were understandable and meaningful
Consistency and standards	Ranking presentation remained consistent across searches
Recognition rather than recall	Users could interpret result cards without memorization
Transparency and trust	Explanations improved trust and understanding

Table 5: Nielsen heuristic evaluation

The heuristic evaluation demonstrates that the proposed prototype aligns well with established usability engineering principles. In particular, visibility of system status, consistency of ranking presentation, and metadata-based explanations contributed positively to usability and transparency.

These findings are consistent with Nielsen's usability engineering framework, where clear feedback and understandable interaction mechanisms improve overall user experience [5].

8 Discussion

The evaluation results indicate that the proposed fuzzy HCIR system successfully improves exploratory dataset retrieval while maintaining transparency and interpretability.

Participants generally rated the system positively across usability, relevance, and explainability dimensions. In particular, the ability to support vague and exploratory queries aligns with previous research in Human-Centered Information Retrieval and exploratory search [4].

The findings also support prior literature on explainable information retrieval systems, where transparent ranking explanations improve user trust and interpretability [1].

Compared to traditional keyword-based retrieval systems, the proposed fuzzy logic approach provides more flexible and interpretable relevance estimation through gradual multi-criteria reasoning.

The integration of metadata-driven explanations significantly improved users' understanding of ranking behavior. This observation aligns with previous work in explainable AI systems, where understandable explanations improve trust and usability.

Furthermore, the use of fuzzy inference enabled the system to combine multiple metadata features without relying solely on strict keyword matching. This improved robustness in exploratory search scenarios.

8.1 Comparison with Existing Literature

The findings of this evaluation are consistent with previous research in Human-Centered Information Retrieval and exploratory search systems [4]. Similar studies have demonstrated that interactive retrieval interfaces and transparent ranking explanations improve user engagement and trust.

The results also support prior work in explainable information retrieval and interpretable AI systems, where understandable ranking explanations contribute to improved interpretability and transparency [1].

Furthermore, the effectiveness of gradual multi-criteria relevance estimation observed in this evaluation aligns with previous fuzzy information retrieval research [3], where fuzzy reasoning improves retrieval flexibility for vague and exploratory queries.

8.2 Implications for Human-Centered Information Retrieval

The evaluation results demonstrate that combining fuzzy inference with explainable ranking mechanisms can significantly improve exploratory dataset retrieval in Open Government Data environments.

Unlike traditional keyword-based retrieval systems, the proposed HCIR framework supports gradual relevance estimation and more flexible interaction with vague or incomplete queries. This is particularly important in exploratory search scenarios, where users may not have clearly defined information needs at the beginning of the search process.

The integration of metadata-driven explanations also improved interpretability and transparency. Participants reported that explanation mechanisms helped them better understand ranking behavior and increased confidence in retrieval results. These findings support previous research emphasizing the importance of explainability and transparency in Human-Centered Information Retrieval systems [4, 1].

Furthermore, the fuzzy inference framework enabled the retrieval system to combine multiple metadata attributes simultaneously without relying solely on exact keyword matching. This improved the robustness of ranking behavior for exploratory search tasks and reduced the limitations commonly associated with strict Boolean retrieval models.

The evaluation additionally demonstrates that explainability and usability are closely connected in exploratory retrieval systems. Transparent ranking explanations not only improved interpretability but also supported user decision-making during dataset selection.

9 System Refinements After Evaluation

Following the Design Science Research methodology, several refinements were implemented based on user feedback and evaluation observations.

The following improvements were introduced:

- Improved title handling for multilingual datasets

- Removal of invalid or empty result entries
- Improved ranking visualization
- Enhanced metadata presentation
- Cleaner result card rendering
- Improved handling of vague exploratory queries
- Better result filtering

The evaluation process also helped identify rendering issues associated with incomplete metadata fields. These issues were resolved through additional validation and filtering mechanisms.

These refinements improved system robustness and usability without requiring major architectural modifications.

The refinement process follows the iterative Design Science Research methodology, where evaluation feedback is continuously used to improve artifact quality, usability, and robustness [2].

Issue Identified	Refinement Implemented
Invalid result rendering	Added validation filtering
Empty multilingual titles	Added safe title handling
Visual clutter in result cards	Simplified layout presentation
Broad exploratory queries	Improved query handling logic
Metadata inconsistencies	Added metadata normalization

Table 6: System refinements after evaluation

The refinement process demonstrates the iterative nature of Design Science Research, where evaluation findings are continuously used to improve artifact quality and usability. Most refinements focused on improving robustness, metadata presentation, and exploratory search interaction without requiring major architectural changes.

10 Limitations

Despite positive evaluation results, several limitations remain.

First, the user evaluation involved a relatively small number of participants. Larger-scale studies would improve the generalizability of the findings.

Second, the evaluation focused primarily on metadata-driven retrieval and did not integrate advanced semantic embedding models such as Sentence-BERT.

Third, the fuzzy rule base still requires manual construction and tuning.

Finally, the prototype currently focuses primarily on metadata interpretation rather than deeper semantic understanding of dataset content.

Future work could integrate:

- semantic embedding models

- adaptive fuzzy rule learning
- multilingual semantic retrieval
- larger-scale evaluation datasets

Another limitation of the current evaluation is the relatively homogeneous participant group, which primarily consisted of Master's-level students. Although the participants were familiar with dataset retrieval tasks, a more diverse evaluation involving researchers, public-sector employees, and domain experts would improve the generalizability of the findings.

In addition, the evaluation relied partially on subjective relevance judgments collected through questionnaire-based feedback. While user-centered evaluation is important in Human-Centered Information Retrieval systems, subjective assessments may vary depending on participants' search experience and familiarity with Open Government Data platforms.

The current prototype also focuses primarily on metadata-driven retrieval and explainability. Although metadata-based ranking improves interpretability and transparency, deeper semantic understanding of dataset content remains limited. Integrating semantic embedding models such as Sentence-BERT could improve semantic retrieval capabilities in future work.

Finally, the fuzzy inference framework currently requires manual rule construction and parameter tuning. Although this improves interpretability and control over the ranking process, automatic fuzzy rule learning mechanisms could improve scalability and adaptability in larger retrieval environments.

11 Conclusion

This phase evaluated the proposed fuzzy Human-Centered Information Retrieval system using both quantitative and qualitative approaches.

The evaluation demonstrated that the system effectively supports exploratory dataset retrieval while maintaining transparency and interpretability.

The fuzzy logic-based ranking mechanism enabled flexible multi-criteria relevance estimation, while the generated explanations improved user trust and understanding.

The results indicate that combining fuzzy inference with metadata-driven retrieval improves both usability and interpretability in Open Government Data systems.

Furthermore, the iterative refinement process aligned with Design Science Research principles and contributed to improving system robustness and user experience.

The evaluation results support previous research emphasizing the importance of explainability, transparency, and exploratory interaction in Human-Centered Information Retrieval systems [4, 1].

Future work may extend the system using semantic retrieval techniques and larger-scale evaluations.

A Evaluation Questionnaire

The evaluation questionnaire used during the user-centered evaluation consisted of usability, relevance, explainability, and satisfaction-related questions based on a 5-point Likert scale.

The following scale was used:

- 1 = Strongly disagree
- 2 = Disagree
- 3 = Neutral
- 4 = Agree
- 5 = Strongly agree

Example questionnaire items included:

- The interface is easy to understand.
- The search process felt intuitive.
- The returned datasets were relevant to my search queries.
- The explanations for ranking decisions were understandable.
- The relevance scores improved transparency and trust.

B Example Search Queries

The following exploratory queries were used during evaluation:

- transport datasets
- climate statistics
- healthcare data
- environmental monitoring
- population statistics
- public mobility data

Participants were also encouraged to perform free exploratory searches using topics of personal interest.

C Example Generated Explanations

The prototype generated metadata-driven explanations describing why datasets were ranked highly.

Example explanation:

> "This dataset was ranked highly because it closely matched the query terms, contained complete metadata, and was recently updated."

Another example explanation:

> "The dataset received a high relevance score because of strong metadata completeness, keyword similarity, and recent publication activity."

D Example User Feedback

The following comments were collected during the evaluation process:

- "The explanations helped me understand why datasets were ranked highly."
- "The system handled vague search queries effectively."
- "Metadata presentation improved trust in the search results."
- "The interface was easy to use and visually clear."

References

[1] Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*, 2017.

[2] Alan R. Hevner, Salvatore T. March, Jinsoo Park, and Sudha Ram. Design science in information systems research. *MIS Quarterly*, 28(1):75–105, 2004.

[3] Donald H. Kraft and Carolyn Colvin. *Fuzzy Information Retrieval*. Springer, 2017.

[4] Gary Marchionini. Exploratory search: From finding to understanding. *Communications of the ACM*, 49(4):41–46, 2006.

[5] Jakob Nielsen. *Usability Engineering*. Morgan Kaufmann, 1994.